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Dual-reference approach for vision-based structural displacement measurement using time-varying Kalman filter

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Abstract

Accurate displacement measurement is essential for structural health monitoring (SHM) to ensure infrastructure safety. Most previous vision-based displacement measurement methods either rely on static reference frames or lack dynamic error feedback, leading to performance degradation under real-world conditions. To address these challenges, this study proposes the dual-reference Kanade–Lucas–Tomasi (DR-KLT) method, which improves vision-based displacement measurements by dynamically integrating both the initial reference frame and the previous reference frame in the KLT tracker. The proposed method estimates the reliability of tracking by analyzing performance indicators such as corner tendency, bi-directional error, number of feature points, and optical flow magnitude. These estimates are incorporated into a time-varying Kalman filter for accurate displacement estimation. Validation through simulations, lab-scale, and on-site experiments demonstrate the method's robustness and superior accuracy compared to single-reference approaches. The results confirm that the DR-KLT approach effectively mitigates the limitations of conventional KLT-based tracking under unstable conditions such as occlusion or lighting variation, making it a reliable tool for real-world SHM applications.

1 | INTRODUCTION

The deterioration of aging infrastructure is a growing concern, as illustrated by numerous reports and news articles in recent years. The safety and functionality of aging structures have come under scrutiny due to several catastrophic failures. For example, in 2007, the I-35 Bridge over the Mississippi River in Minneapolis collapsed after 40 years in service, resulting in 13 deaths and 145 injuries (Salem & Helmy, 2014). Similarly, in 2018, the partial collapse of the Morandi Bridge in Genoa, Italy, 51 years after its construction, caused 43 fatalities (Milillo et al., 2019; Morgese et al., 2020). These tragic incidents underscore the urgent need for efficient and reliable methods to monitor, assess, and maintain aging infrastructure to prevent future disasters.

A prominent method for assessing the condition of structures is vibration-based structural health monitoring (SHM), which involves collecting dynamic data and analyzing it in the frequency domain to assess the structure's condition (Beck & Jennings, 1980; Safak, 1991; Şafak, 1989; Soyoz & Feng, 2008; Amezquita-Sanchez & Adeli, 2016; Tsogka et al., 2017, 2019; Razavi et al., 2024). Productivity and safety monitoring in building construction have been significantly advanced by integrating ultra-wideband positioning and range imaging technologies (Altunok et al., 2006; Teizer & Castro-Lacouture, 2007; Xianjia et al., 2021). These advancements lay the foundation for extending monitoring techniques to structural vibration measurement. The application of wireless sensor networks in SHM has been explored extensively, as demonstrated by studies like the development of a web-based building environmental monitoring system (Jang et al., 2008; Sofi et al., 2022). Effective SHM requires prior data collection, traditionally performed by attaching contact sensors directly to the structure and connecting them with wires (Javadinasab Hormozabad et al., 2021; Tan et al., 2011). Traditionally, data collection for structural vibration involves using accelerometers and displacement sensors (Abdulkarem et al., 2020). Bedon et al. (2018), Amezquita-Sanchez et al. (2018), and Park et al. (2007) utilized a self-developed MEMS (micro electro-mechanical system-based) accelerometer to collect data and applied experimental modal analysis (EMA) methods to determine the dynamic characteristics of structures. Li et al. (2015) developed sensors to measure the relative displacement between the slab and girder, monitoring the damage of shear connectors to validate the sensors. Although contact sensors have been used from the beginning, their installation is not straightforward, often requiring significant time and cost. The advent of wireless sensors in the early 2000s facilitated data collection (Bocca et al., 2011; Hu et al., 2013). Zhu et al. (2018) and Nagayama & Spencer, 2007 developed a high-sensitivity wireless accelerometer based on the Xnode platform to accurately measure vibrations in structures and analyze these data to obtain essential information for monitoring civil infrastructure. Noncontact sensors such as laser displacement meters have been developed, but these either lack accuracy or require high costs for high precision.

Recently, vision-based measurement techniques have been increasingly adopted. Compared to traditional sensor-based methods, vision-based methods offer distinct advantages (Feng et al., 2015; Gao et al., 2023; S. Lee, Kim, and Sim, 2022; Ngeljaratan et al., 2021; Sun et al., 2022; Zhang et al., 2024; Spencer et al., 2025). Vision-based methods are easier to install, cost-effective, and can measure multiple points simultaneously. One of the most popular methods for vision-based vibration measurement is optical flow-based methods such as Kanade–Lucas–Tomasi (KLT) Tracker. KLT Tracker detects movements of feature points within the region of interest (ROI) between consecutive images using the intensity differences. Yoon et al. (2016) introduced utilizing KLT tracker for measuring structural vibration using commercial camera, directly extracting feature points of structural components without attaching any artificial targets. Yoon et al. (2018) compensated the motion of the camera so that unmanned aerial vehicles can be used to measure vibration. G. Lee, Kim, Ahn, et al. (2022) calibrated the video recorded from nonperpendicular plan, expending the feasible location of the camera for vision-based measurement. Yan et al. (2022) developed an algorithm combining KLT and Convolutional Neural Network (CNN) and proposed a method to measure displacement based on sequential images obtained from unmmanned aerial vehicle (UAV). Jeon et al. (2022) introduced Uni-KLT, a vision-based method for measuring vibration of cables. Nguyen et al. (2025) proposed a hybrid approach that combines KLT with an machine learning (ML)-based prediction model to improve vision-based structural displacement measurement. Such innovations significantly enhance the ability to monitor structures over extended periods, directly informing maintenance decisions and reducing downtime.

Although KLT tracker have been widely adopted for structural vibration measurements, the results of KLT tracker vary depending on which image frame is selected as the reference image frame. When the immediate previous frame is selected as the reference image frame (previous reference frame, denoted as PRF hereafter), the displacement (residual displacement) between the two image frames can be accurately predicted. However, over time, cumulative errors can lead to a decrease in the accuracy of the displacement relative to the initial position. Conversely, when the initial image frame of the video is selected as the reference frame (initial reference frame, denoted as IRF hereafter), the accuracy of the displacement relative to the initial position can be more accurate. However, the accuracy decreases when large displacements occur. Simulation results demonstrate this problem more clearly: the root mean square error (RMSE) of IRF tracking was measured at 28.64 pixels, while PRF tracking produced an RMSE of 24.75 pixels, highlighting the cumulative drift error that occurs over time. These limitations indicate that neither IRF nor PRF alone is sufficient to guarantee reliable long-term tracking performance. Specifically, significant discrepancies can arise between the ROI and the actual object during long-term monitoring.

This paper proposes a new approach, dual-reference KLT (DR-KLT), which can take advantage of both IRF and PRF. Even within a single video, IRF can provide more accurate results than PRF in certain frames, and PRF can provide more accurate results than IRF in other frames. The proposed method tracks the feature points for each frame using both IRF and PRF and combines the result using time-varying Kalman filter. Although it is difficult to estimate the error of the KLT tracker using IRF and PRF, they can be inferred from the performance indicators during the tracking process. Therefore, the proposed method analyzed the performance indicators that can be obtained during the tracking process (corner tendency, the number of feature points, bi-directional tracking results, and the magnitude of optical flow), and utilized the performance as covariance of the time-varying Kalman filter.

The contributions of this paper are as follows:

1. Analyzed the performance indicators for KLT tracker: (1) corner tendency, (2) the number of feature points, (3) bi-directional tracking results, and (4) the magnitude of optical flow.
2. Developed error estimation model for KLT tracker using XGBoost that can estimate the error of the KLT tracker for each frame using the performance indicators.
3. Developed DR-KLT, which improved the robustness and the estimated error of IRF and PRF.

The remainder of this paper is organized as follows: Section 2 discusses the analysis of the performance indicators during the tracking process of the KLT tracker. Section 3 introduces the error estimation model for the KLT tracker using these performance indicators and presents the DR-KLT method, which combines the KLT tracker with a time-varying Kalman filter. Section 4 presents validation experiments for the proposed method, and finally, Section 5 presents the conclusions of this study.

2 | ANALYSIS OF THE PERFORMANCE INDICATORS FOR KLT TRACKER

To obtain a more accurate state estimation using the Kalman filter it is essential to analyze the uncertainty of the KLT tracker for the given data by accurately estimating the process noise covariance matrix and the measurement noise covariance matrix. The accuracy of the KLT tracker can vary significantly depending on video capture conditions and even between frames within the same video. Therefore, to apply a time-varying Kalman filter that dynamically adapts to these variations, it is necessary to predict the uncertainty of the KLT tracker for each frame. The accuracy of the KLT tracker can fluctuate due to several factors, including image resolution, contrast, the shutter speed used for sequential image capture, and other external conditions. Among these factors, this study identified four key performance indicators that significantly influence the KLT tracker's tracking accuracy.

2.1 | Corner tendency

The KLT tracker usually selects multiple feature points to be tracked, instead of tracking every pixel in the image. To reduce processing time and increase the reliability of data, only a subset of the image is selected and used as feature points (Tomasi & Kanade, 1991). It is generally accepted that corner points serve as reliable feature points for tracking (Shi, 1994).

However, few studies have analyzed the relationship between corner tendency and the performance of KLT trackers (Sheorey et al., 2015). Therefore, this study analyzed how the corner tendency can affect the performance of the KLT tracker.

The structure tensor in an image $I(x, y)$ is defined as follows:

$$G = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \tag{1}$$

where I_x and I_y are the gradients of the image in the x and y directions, respectively.

The eigenvalue of the structure tensor G can be obtained by solving the characteristic equation:

$$\det(G - \lambda I) = 0 \tag{2}$$

where λ represents the eigenvalues and I is the identity matrix. The eigenvalues λ_1 and λ_2 are given by:

$$\lambda_1 = \frac{I_x^2 + I_y^2 + \sqrt{\left(I_x^2 + I_y^2\right)^2 + 4\left(I_x I_y\right)^2}}{2} \tag{3}$$

$$\lambda_2 = \frac{I_x^2 + I_y^2 - \sqrt{\left(I_x^2 + I_y^2\right)^2 + 4\left(I_x I_y\right)^2}}{2} \tag{4}$$

Solving this equation yields the two eigenvalues λ_1 and λ_2. Shi (1994) evaluated points using the smaller of the two eigenvalues λ_1 and λ_2. The corner tendency R is defined as the minimum of the two eigenvalues of G:

$$R = \min(\lambda_1, \lambda_2) \tag{5}$$

R has a minimum value of 0 and, theoretically, can have an infinite maximum value. However, due to the intensity values of the image, R has a finite maximum value. To calculate the theoretical maximum value of R, consider that the pixel intensity values of an image typically range from 0 to X. If $I_x = I_y = X$, the structure tensor G is:

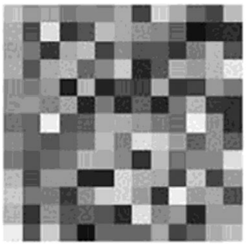

FIGURE 1 Example of targets with various corner tendencies.

$$G = \begin{bmatrix} X^2 & X^2 \\ X^2 & X^2 \end{bmatrix} \tag{6}$$

Calculating the eigenvalues of this matrix gives:

$$\lambda_1 = \lambda_2 = \frac{2 \times X^2 \pm \sqrt{0}}{2} = X^2 \tag{7}$$

Therefore, the corner tendency R is X^2, which is the maximum value within the range of pixel intensity values in an image.

To analyze the relationship between the corner tendency and the performance of the KLT tracker, a simulation test was conducted by using feature points with various corner tendencies. First, a specially designed checkerboard was used to generate corner points with different corner tendencies as shown in Figure 1.

An ROI was set in the area corresponding to the target in the image, and a corner detection algorithm was applied to the ROI to extract corner points. The extracted corner points were divided into three sections based on their corner tendency, ensuring each section had an equal number of corner points for the experiment. The experiment involved moving the target by displacements ranging from 1 to 15 pixels and tracking only the corner points with the corresponding corner tendency. A total of 1500 consecutive images were used for the experiment, with 100 trials conducted for each magnitude of optical flow. The error was calculated by comparing the measured results with the true displacement of the target, and the error according to the corner tendency is shown in Figure 2.

The results indicated that although a higher corner tendency did not consistently result in smaller errors, corner points with higher corner tendencies generally exhibited a narrower error range. This suggests that, despite large motion magnitudes or other disturbances, using feature points with higher corner tendencies enables more stable tracking. Additionally, to examine the correlation between corner tendency and error, the correlation coefficient was calculated. The correlation coefficient value was −.1051, indicating that as the corner tendency increases, the error decreases. This indicates that corner tendency is associated with the accuracy of the KLT tracker. However, as the correlation coefficient does not account for nonlinear relationships, an error prediction model that can consider such nonlinearity is required.

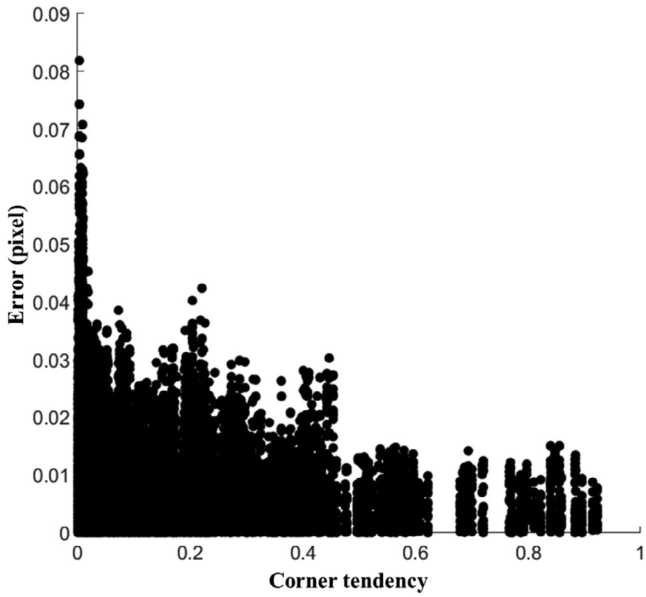

FIGURE 2 Error distribution based on corner tendency.

2.2 | Bi-directional error

The KLT tracker can improve its robustness using bi-directional error detection. Bi-directional error is determined by tracking a feature point in two directions: first forward and then backward. If the feature point tracked forward and re-tracked backward deviates significantly from its original position, the error is deemed excessive, and the point is considered unreliable. Although bi-directional error is typically used to enhance accuracy by eliminating points exceeding a predefined threshold, this study investigates its impact on the performance of the KLT tracker

Figure 3 illustrates the occurrence of bi-directional error through forward and backward tracking within the current frame. Let P be the feature point in the current frame. Through forward tracking, P is matched to the corresponding point P' in the next frame. P' is then tracked back to its corresponding point P'' in the current frame using backward tracking.

$$E_{bi} = \|P - P''\|_2 \tag{8}$$

E_{bi} represents the bi-directional error, and generally, a larger E_{bi} indicates lower confidence in the tracking accuracy. It is defined as $E_{bi} = \|P - P''\|_2$, where $\| \|_2$ denotes the Euclidean norm (L2 norm). This clarification has been added to avoid ambiguity. The bi-directional error is

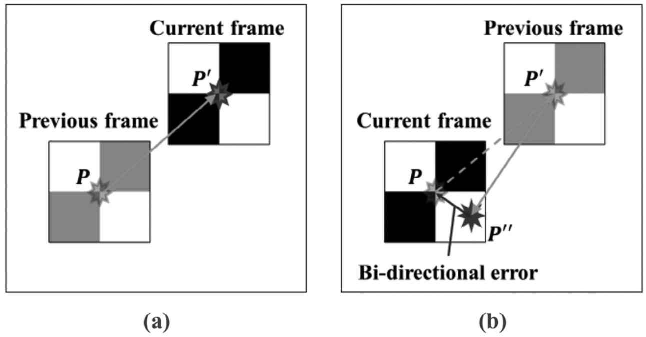

FIGURE 3 Illustration of bi-directional error: (a) forward tracking; (b) forward and backward tracking.

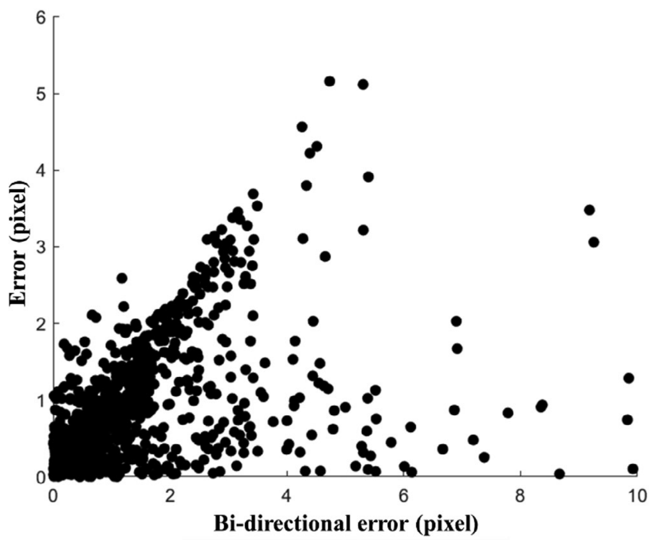

FIGURE 4 Error distribution based on bi-directional error.

influenced by various factors, which makes it challenging to achieve a uniform distribution of values. To generate a wide range of bi-directional error values, artificial noise was introduced into the images.

The error corresponding to the measured bi-directional error is shown in Figure 4. As bi-directional error is influenced by multiple factors, it is difficult to isolate its effects from other variables. Nevertheless, the results indicate that tracking error increases proportionally with bi-directional error. The correlation coefficient value was .3522, indicating a positive relationship between bi-directional error and tracking error. This suggests that bi-directional error is significantly correlated with the accuracy of the KLT tracker.

2.3 | Number of feature points

The KLT tracker calculates a transformation matrix based on the motion of feature points and applies it to estimate the object's motion. During this process, the type of transformation varies depending on the nature of the object's motion, and consequently, the required minimum number of feature points also changes (Table 1). For example, if the object exhibits in-plane body motion and it is assumed that the camera is parallel to the structure, a rigid body motion is characterized by all feature points moving in the same direction and with the same magnitude. In this case, tracking can be achieved with a minimum of one feature point. However, if out-of-plane motion occurs and the camera is not parallel to the object, the angles, lengths, and parallelism of the objects within the frame must be considered. A projective transformation can accommodate such complex motions, and it requires at least four feature points.

TABLE 1 Minimum number of feature points required for each transformation type.

Transformation type	Minimum number of feature points
Translation	1
Rigid	2
Similarity	2
Affine	3
Projective	4

Furthermore, the KLT tracker employs the Maximum Likelihood Estimation Sample Consensus (MLESAC) algorithm during the calculation of the transformation matrix to eliminate outliers and achieve more accurate results. When the number of given feature points exceeds the required minimum, a single transformation matrix cannot adequately describe all the feature points. Therefore, the transformation matrix is estimated based on a random subset of feature points, and the error and corresponding likelihood cost between the transformation matrix and all feature points are calculated. The transformation matrix with the lowest likelihood cost is chosen as the optimal solution. This process is defined as MLESAC. Feature points with deviations exceeding the threshold from the optimal transformation matrix are excluded from the tracking process, resulting in improved accuracy.

To analyze the correlation between the number of feature points and the performance of the KLT tracker, a simulation video was created with a moving object. Corner points of the target object were extracted from the reference frame of the video, and a random subset of these extracted corner points were selected. As the simulation assumes a rigid transformation type where the shape of the object does not change, the number of selected feature points was set to be two or more. The tracking process was conducted using the selected points, and the corresponding errors are presented in Figure 5.

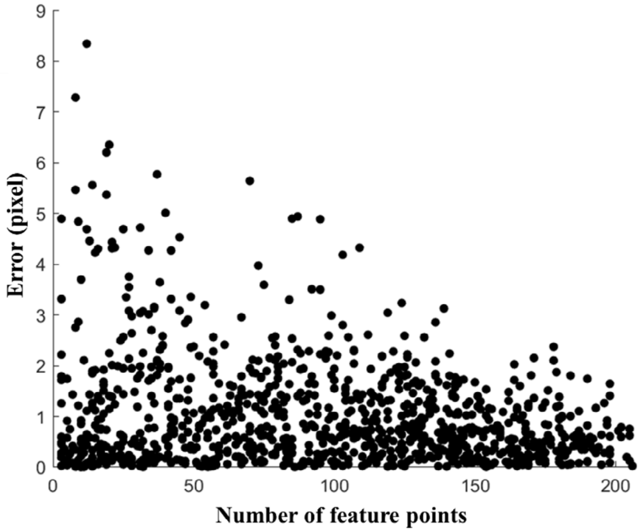

FIGURE 5 Error distribution according to the number of feature points.

The number of randomly selected feature points ranged from 2 to 210, and as this number increased, the maximum error generally decreased. This indicates that while accurate results can be obtained with a smaller number of feature points, a larger number of feature points is necessary to consistently achieve good results. The coefficient value was –.2385, indicating a negative correlation between the number of feature points and the tracking error. Reliable tracking results can still be achieved even with a small number of feature points. However, a larger number of feature points consistently yielded more stable and accurate tracking performance.

2.4 | Magnitude of optical flow

Tracking algorithms operate under the assumption that sequential images taken at adjacent time intervals will exhibit minimal changes in the object's motion, shape, and intensity (Lucas & Kanade, 1981). This assumption enables the use of gradient-based methods, which are computationally efficient and depend on small adjustments to the tracked points. When inter-frame changes are minimal, the optical flow can be accurately estimated using local approximations. Therefore, in this study, the validity of this assumption is examined, and the influence of optical flow magnitude on the KLT tracker's performance is analyzed.

The intensity $I(x, y, t)$ of a pixel at coordinates (x, y) at time t is assumed to be equal to the intensity at coordinates $(x + u, y + v)$ at the next time step $t + \Delta t$. Mathematically, this relationship is expressed as:

$$I(x, y, t) = I(x + u, y + v, t + \Delta t) \tag{8}$$

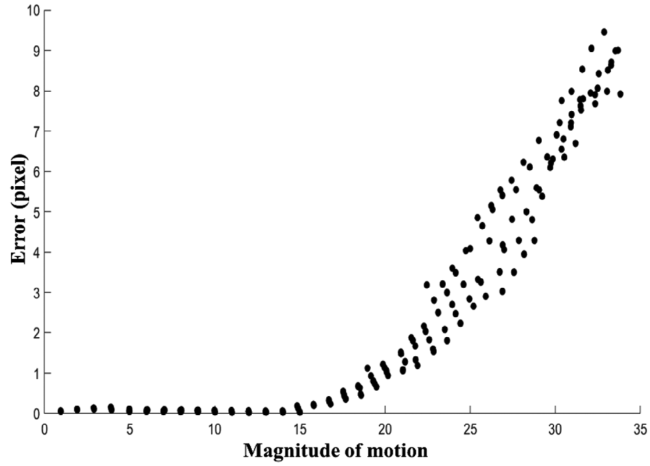

FIGURE 6 Error distribution based on the magnitude of the optical flow.

where u and v are the magnitudes of optical flow in the x and y directions, respectively. If u and v are assumed to be sufficiently small, the equation can be expanded using a Taylor series and approximate to its first-order term as follows:

$$\begin{aligned} &I(x+u, y+v, t+\Delta t) \\ &\approx I(x, y, t) + \frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v + \frac{\partial I}{\partial t}\Delta t \end{aligned} \tag{9}$$

Subtracting $I(x, y, t)$ from both sides results in:

$$\frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v + \frac{\partial I}{\partial t}\Delta t \approx 0 \tag{10}$$

Thus, it can be approximated that:

$$\frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v \approx -\frac{\partial I}{\partial t}\Delta t \tag{11}$$

This equation is valid only when the movement u and v is small. If the movement is large, the higher order terms in the Taylor series cannot be ignored, and the linear approximation will not hold, resulting in decreased tracking accuracy.

To compare the magnitude of optical flow, the displacement between adjacent sequential images was controlled to validate tracking performance. A simulation experiment was configured where only the target was moved by a constant amount between two similar images. The target's displacement varied from 0 to 10 pixels, and each displacement was tracked at different positions and directions, with the experiment repeated 1000 times. This resulted in the outcomes shown in Figure 6, where the x-axis represents the motion measured through the KLT tracker.

The experimental results indicate that as the magnitude of optical flow increases, the range of tracking errors also expands. This reinforces the need for the KLT tracker's

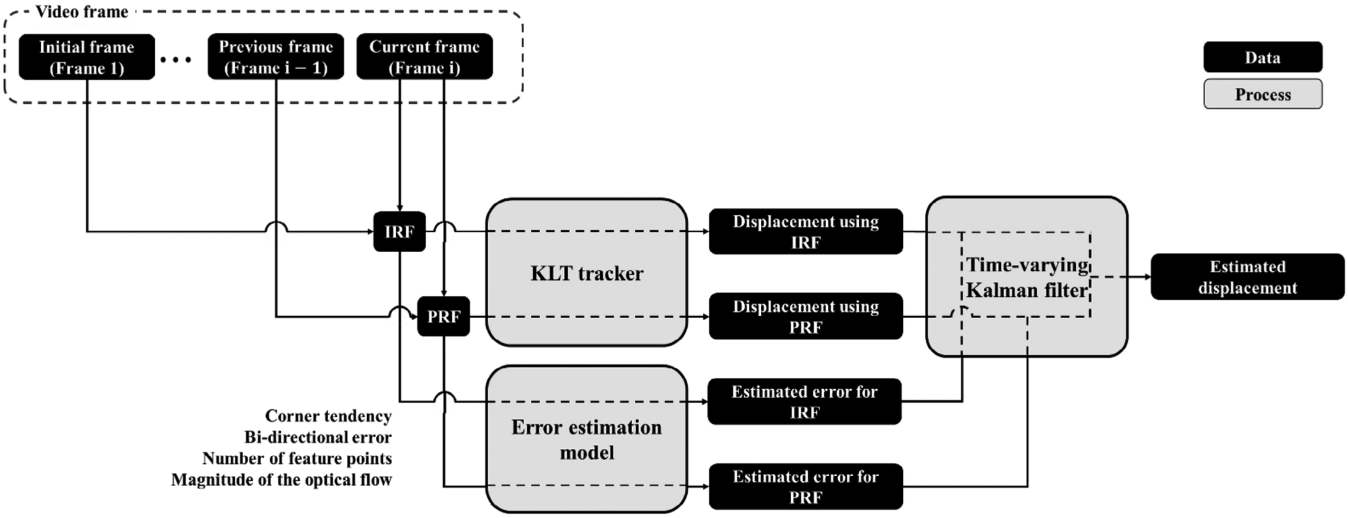

FIGURE 7 Flowchart for the proposed DR-KLT.

assumption that objects undergo very small changes in motion between consecutive image frames. Additionally, the correlation coefficient was .8686, suggesting that the observed relationship may be influenced by nonlinear factors that cannot be fully captured by the linear correlation coefficient. Therefore, to reflect such characteristics in the error prediction model, it is necessary to use a model capable of considering nonlinear relationships.

3 | SYSTEM DEVELOPMENT

This paper aims to achieve higher stability in displacement measurements by integrating two different approaches for implementing the KLT tracker: (1) selecting the initial frame as the reference frame (IRF), and (2) selecting the previous frame as the reference frame (PRF). To integrate the two approaches, a time-varying Kalman filter was employed, and the performance indicators analyzed in Section 2 were used to estimate the error for each of the two approaches in every frame.

Based on these indicators, a model was developed to predict errors, which were then utilized as uncertainty in the input data for the time-varying Kalman filter. This approach facilitates the generation of a unified displacement dataset by merging the two individual datasets. The flowchart illustrating the proposed method is shown in Figure 7.

3.1 | Dual-reference KLT

Generally, KLT operates on consecutive image frames, under the assumption of minimal changes in object motion and appearance between them. This assumption allows for using the previous frame as the reference frame, enabling reliable computation of the optical flow through local approximations. However, this approach results in error accumulation over time, particularly as the length of the video sequence increases. Using the initial frame as the reference can mitigate the accumulation of such errors; however, if the difference between the initial and current frame becomes too large, significant tracking errors may still occur.

3.1.1 | KLT

The KLT tracker operates by minimizing the sum of squared differences (SSD) between pixel intensities in two consecutive image frames. The core principle is to compute the optical flow under the assumption that inter-frame motion is small and can be linearly approximated. As described earlier in Equations (8)–(10), the algorithm assumes brightness constancy and applies a first-order Taylor expansion to derive the optical flow constraint equation.

This constraint is then solved by minimizing the SSD over a window W of neighboring pixels to estimate the motion vectors (u, v) for each feature point.

$$A \cdot d = b \tag{12}$$

where:

$$A = \sum_{p \in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, b = -\sum_{p \in W} \begin{bmatrix} I_x I_t \\ I_y I_t \end{bmatrix} \tag{13}$$

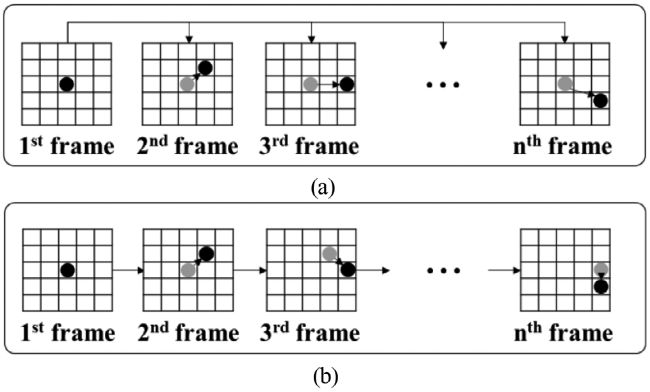

FIGURE 8 The difference when using the initial frame and the previous frame: (a) when the initial frame is selected as the reference frame (IRF); (b) when the previous frame is selected as the reference frame (PRF).

Here, I_x, I_y, and I_t are the partial derivatives of the image with respect to x, y and t (time), respectively. Solving this linear system gives the displacement vector $d = [u, v]^T$:

$$d = A^{-1} \cdot b \tag{14}$$

This procedure allows the KLT tracker to estimate the motion between two frames by minimizing tracking errors through least squares optimization. The displacement d provides the optical flow, which is used to track feature points across consecutive frames.

3.1.2 | Dual reference frames

Tracking algorithms detects small movements by analyzing sequential images captured at closely spaced time intervals. Most tracking algorithms use the image from the previous time step as the reference frame. This approach enhances computational efficiency while maintaining high tracking accuracy. However, tracking errors accumulate as the number of time steps increases. Using the initial image frame as a reference can mitigate cumulative errors; however, as the displacement between the initial and current frames becomes larger, tracking accuracy may decrease. In this study, both the initial frame and the previous frame are employed as reference points to evaluate tracking performance. Figure 8 illustrates the difference between using the initial frame and the previous frame as reference points.

Each reference frame offers distinct advantages and disadvantages. For a frame at time t, let p_{t-1} denote the object's position in the previous frame and p_t denote its position in the current frame. The displacement at time t, denoted as d_t is calculated as:

$$d_t = p_t - p_{t-1} \tag{15}$$

However, errors accumulate over time such as:

$$E_t = \sum_{i=1}^{t-1} e_i \tag{16}$$

Conversely, using the initial frame results in displacements are defined as:

$$d_t = p_t - p_0 \tag{17}$$

This prevents cumulative errors; however, as time progresses, the displacement increases proportionally ($|d_t| \propto t$), potentially leading to larger tracking errors. Therefore, a trade-off exists between minimizing short-term errors and avoiding long-term error accumulation.

3.2 | Error estimation model using the performance indicators

The primary objective of this model is to establish a mathematical framework that accurately predicts errors based on the performance indicators. These predicted errors are then integrated into the data fusion process using a time-varying Kalman filter. Accurate error estimation is crucial as it directly affects the reliability of the fused data, ensuring that it closely represents the actual structural vibrations. In this study, various machine learning models were utilized to predict errors based on the performance indicators, and the model that demonstrated the best predictive performance was selected. These error estimation models were developed using machine learning algorithms, which are effective for capturing complex and nonlinear relationships between performance indicators and errors.

3.2.1 | Data composition

Machine learning techniques are employed to identify patterns in data, which are then utilized to predict outcomes in new datasets. To build a robust machine learning model and achieve accurate predictions, an adequate amount of high-quality data that distinctly reveals these patterns is essential. Additionally, having diverse values in the data helps uncover underlying rules. To meet these requirements, a simulation video with a moving target was generated, with noise and blur intentionally introduced to simulate realistic conditions.

TABLE 2 RMSE (root mean square error) of the error estimation model.

Estimation model	RMSE
XGBoost	0.4329
CatBoost	0.4343
Ensemble (Bagging)	0.6777
TabNet	0.4516
TabTransformer	0.5979
A simple neural network with a self-attention module	0.7425

In the simulation, four performance indicators were extracted while tracking the moving object using the KLT tracker. The tracker's results were then compared with the actual movement distances from the simulation to calculate the tracking error. As a result, a dataset with 1000 data points was constructed, each containing the four performance indicators and the corresponding errors. Of the 1000 data points, 800 were used as training data, and the remaining 200 were reserved as test data to evaluate the model's performance.

3.2.2 | Error estimation model selection

The most effective approach to predicting errors from performance indicators involves using machine learning algorithms, which model the relationship between performance indicators and errors through regression analysis. Various machine learning models, such as linear regression, decision tree regression, and gradient boosting machines, are suitable for this analysis. Recently, transformer models incorporating attention mechanisms have become a focal point of research. In this study, the constructed dataset was applied to models optimized for tabular data to select the most efficient and accurate algorithm. The selected models include three based on tree boosting algorithms and three incorporating a self-attention mechanism. For the tree boosting models, hyperparameters including the number of trees, maximum depth, learning rate, and subsample size were fine-tuned. Similarly, for the models utilizing self-attention mechanisms, hyperparameters such as the number of attention heads, dropout rate, and learning rate were adjusted.

Table 2 presents the results of predicting errors from performance indicators when training and testing the dataset proposed in Subsection 3.2.1. The simulation consisted of 300 frames in which the target moved along a sinusoidal path. The average pixel displacement between frames was approximately 3.5 pixels, with a maximum displacement of around 15 pixels from the initial position. For the same dataset, models based on boosting algorithms, such as XGBoost and CatBoost, achieved the highest accuracy, while TabNet, which integrates self-attention with decision trees, also demonstrated competitive performance. For tabular data, models incorporating decision trees, such as those based on boosting algorithms, exhibited relatively better performance. Consequently, the XGBoost model, which exhibited the best performance in this study, was selected for error prediction.

To further improve model performance, we conducted hyperparameter optimization for the XGBoost model using GridSearchCV with fivefold cross-validation. The search space included the following parameters:

Max depth = [3, 4, 5], learning rate = [0.01, 0.1, 0.2], n estimators = [100, 200, 300], subsampling ratio = [0.8, 1.0], and feature subsampling ratio per tree (colsample_bytree) = [0.8, 1.0]. The optimal configuration was selected based on the lowest RMSE on the validation set. The predicted tracking error from the trained XGBoost model was then used to construct the measurement noise covariance matrix R in a time-varying Kalman filter, enabling confidence-adaptive fusion of IRF and PRF tracking results.

3.3 | Data fusion using time-varying Kalman filter

The KLT tracker is an image tracking algorithm designed to detect small changes between consecutive frames, allowing for accurate object tracking within a video sequence. This approach reduces computational complexity and ensures precise tracking by extracting and utilizing feature points from the object in the image. However, the accuracy of the KLT tracker is affected by several factors, such as the tendency of feature point corners, bidirectional error, the number of feature points, and the magnitude of optical flow.

Typically, the KLT tracker uses the image from the previous time step as the reference frame. However, cumulative errors can occur as the tracking process progresses. To mitigate these errors, an alternative approach is to use the initial image frame as the reference. However, this method may experience a loss of accuracy if the object undergoes large displacements. To address these issues, a time-varying Kalman filter is utilized to measure displacement from two reference frames. By integrating these measurements through the Kalman filter, the objective is to obtain more accurate and stable displacement measurements.

The time-varying Kalman filter is an algorithm used to estimate the states of systems with time-varying dynamics. It is highly effective in environments characterized by uncertainty and noise. The Kalman filter utilizes linear algebra and probability theory to estimate the current state of the system and updates this estimate using measurement data.

In this paper, a Kalman filter was designed to integrate two sets of tracked data into a single, refined displacement estimate. The displacement is calculated using the initial image as the reference, whereas the velocity is calculated using the previous image, and these two results are integrated into a unified system. The state vector x of this system is as follows:

$$x = \begin{bmatrix} d \\ v \end{bmatrix} \tag{18}$$

d and v represent displacement and velocity. The data obtained using IRF are used as displacement, while the data obtained using PRF are used as velocity. The state transition matrix is defined as:

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \tag{19}$$

where Δt is the timestep between measurements. The estimated state vector $\hat{x}$ is as follows:

$$\hat{x} = F\hat{x}_{t-1} \tag{20}$$

The state covariance matrix P^{-} represents the uncertainty in the state estimate, and the predicted P^{-} is given by:

$$P^{-} = FP_{t-1}F^{T} + Q \tag{21}$$

where Q is process noise covariance matrix. In this study, Q was determined empirically through parameter tuning to reflect the uncertainty in the system model, following common practice in Kalman filter-based tracking when exact process noise statistics are unavailable. The measurement matrix H maps the internal state vector to the measurement space. As both displacement and velocity are considered measurable outputs in this study, H is defined as the identity matrix:

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{22}$$

The Kalman gain K dictates the extent to which the estimated state should be adjusted based on the measurement error. The Kalman gain K for displacement is as follows:

$$K = P^{-}H^{T}\left(HP^{-}H^{T} + R\right)^{-1} \tag{23}$$

In a typical Kalman filter, the measurement noise covariance matrix R is constructed using the statistical variance (Var) and covariance (Cov) of measurement errors, which reflect the uncertainty in the sensor observations.

$$\begin{aligned} R &= \begin{bmatrix} \mathrm{Var}\left(\mathrm{error}_{d}\right) & \mathrm{Cov}\left(\mathrm{error}_{d}, \mathrm{error}_{v}\right) \\ \mathrm{Cov}\left(\mathrm{error}_{v}, \mathrm{error}_{d}\right) & \mathrm{Var}\left(\mathrm{error}_{v}\right) \end{bmatrix} \\ &= \begin{bmatrix} \frac{\mathrm{Dev}(\mathrm{error}_{d})^{2}}{n} & \frac{\mathrm{Dev}(\mathrm{error}_{d})\cdot \mathrm{Dev}(\mathrm{error}_{v})}{n} \\ \frac{\mathrm{Dev}(\mathrm{error}_{v})\cdot \mathrm{Dev}(\mathrm{error}_{d})}{n} & \frac{\mathrm{Dev}(\mathrm{error}_{v})^{2}}{n} \end{bmatrix} \\ &\approx \begin{bmatrix} \mathrm{error}_{d}^{2} & \mathrm{error}_{d}\cdot \mathrm{error}_{v} \\ \mathrm{error}_{v}\cdot \mathrm{error}_{d} & \mathrm{error}_{d}^{2} \end{bmatrix} \end{aligned} \tag{24}$$

This statistical approach requires enough samples to compute reliable variance–covariance estimates. However, in real-time applications where such sample sets may not be available, the variance and covariance can be approximated using the estimated instantaneous errors for displacement and velocity at each timestep. To avoid overestimating correlation between the two errors, the covariance matrix is simplified using a diagonal form that assumes independence:

$$R = \begin{bmatrix} \mathrm{error}_{d}^{2} & 0 \\ 0 & \mathrm{error}_{v}^{2} \end{bmatrix} \tag{25}$$

The state covariance matrix P is also updated to reflect the reduced uncertainty after incorporating the measurement:

$$P = (I - KH)P^{-} \tag{26}$$

By applying these estimates at each time step in the sequence, a single fused data estimate is obtained that considers the uncertainties of the measurements based on two reference frames.

4 | VALIDATION

In this paper, the key factors affecting the performance of the KLT tracker were identified, and a method was proposed to integrate KLT and ref-KLT using a time-varying Kalman filter. To achieve this, relevant factors influencing tracking accuracy were selected from sequential image information. The error variations associated with each factor were then analyzed to develop a nonlinear estimation model. This model was utilized to estimate the measurement covariance in the Time-Varying Kalman Filter, and the data obtained from both KLT and ref-KLT were integrated into a unified dataset.

To validate the proposed method, three different types of validation tests were conducted. First, simulations were performed to verify the performance of the proposed method in a controlled environment. Next, to evaluate the method using real-world data, the results were compared with data collected from sensors mounted on an experimental structure subjected to external excitation. Finally, to assess the field applicability of the proposed method, video data from an actual bridge was captured and the technique was applied.

4.1 | Simulation test

A simulation video was generated to validate the performance of the proposed method under controlled conditions. The simulation video used in this test had a resolution of 540 × 540 pixels. A 3-degree-of-freedom shear building with specified material properties was modeled, and random vibrations were applied to generate a response simulation. Each video was 10 s long, consisting of 300 frames at 30 FPS. The simulation controlled the displacement based on the applied magnitude and the structural characteristics. Additionally, various noise and blur effects were applied to each frame to induce a range of tracking errors, enabling the simulation of different scenarios.

When using PRF, better results are obtained when cumulative displacement is large and environmental factors such as noise are minimal. Conversely, when cumulative displacement is small but environmental factors like noise are significant, IRF yields more accurate results. Additionally, as the duration of the video (total number of frames) increases, errors accumulate when using PRF, while IRF remains unaffected by such cumulative errors. To verify the accuracy of the proposed method under these conditions, simulations were conducted considering both scenarios. For this purpose, various experimental cases were generated by adjusting the maximum possible displacement, the duration of the video, and levels of blur and noise. To illustrate an example of the simulation data, Figure 9 presents the error trends observed in Case 8. As shown, PRF exhibits oscillatory behavior due to cumulative errors, where the error magnitude increases and then decreases over time. In contrast, IRF demonstrates relatively stable fluctuations within a bounded range. These trends highlight the distinct error characteristics of each method under the same conditions. The simulations were designed to represent scenarios that could realistically occur in railway bridges, and the quantitative results from these simulations are presented in Table 3.

Table 3 shows the average performance indicators and the RMSE of displacement measurements for each method. In Cases 1, 2, 4, and 8, using the IRF results in larger bi-directional errors, detects fewer feature points, and shows slightly higher magnitudes of optical flow compared to the PRF. Consequently, the error prediction model

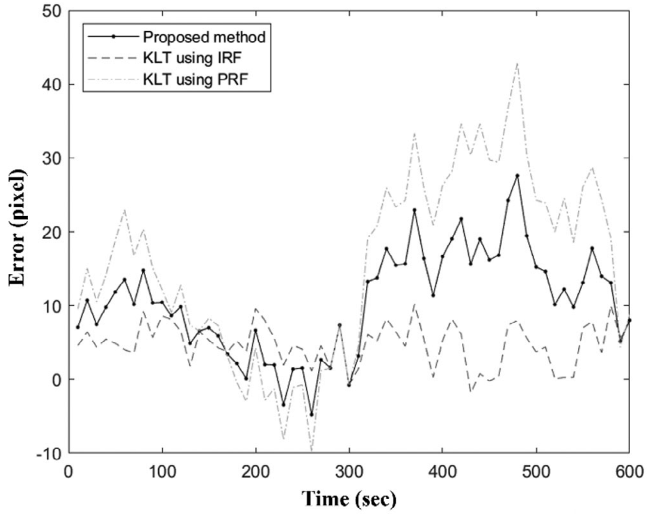

FIGURE 9 Average tracking error computed over 300-sample intervals for visual clarity.

TABLE 3 RMSE (root mean square error) of the error estimation model.

Case	Corner tendency (%)		Bi-directional error (pixel)		Number of feature points (count)	
	IRF	PRF	IRF	PRF	IRF	PRF
1	0.45	0.44	221.01	3.21	52.00	62.33
2	0.45	0.43	138.86	8.06	52.00	130.46
3	0.01	0.36	30.35	4.92	68.00	63.14
4	0.45	0.37	137.82	5.35	52.00	74.82
5	0.04	0.43	3.14	7.48	59.00	42.12
6	0.45	0.58	2.70	8.29	52.00	74.71
7	0.04	0.36	30.02	5.34	59.00	72.73
8	0.45	0.50	30.30	5.77	52.00	102.01
Case	**Magnitude of optical flow (pixel)**		**RMSE (pixel)**			
	IRF	**PRF**	**IRF**	**PRF**	**Kalman filter**	**DR-KLT**
1	2.42	0.12	67.40	48.87	48.88	50.83
2	2.95	0.05	67.95	34.30	34.31	38.86
3	9.85	0.05	6.89	8.75	8.75	7.98
4	2.83	0.00	67.60	70.27	70.27	69.78
5	1.74	0.02	3.09	3.64	3.64	3.06
6	1.94	0.01	6.26	7.38	7.37	6.73
7	0.15	0.01	4.71	4.71	4.71	4.68
8	0.11	0.01	5.19	20.09	19.97	12.64
Average			28.64	24.75	24.74	24.32

indicates higher uncertainty for the IRF, whereas the DR-KLT yields results closer to those of the PRF.

In Case 3, despite detecting more feature points with the IRF, the low corner tendency, high bi-directional error, and higher optical flow magnitude lead to greater predicted uncertainty, resulting in inaccurate outcomes even though IRF accuracy was initially higher. In Case 6, the PRF was expected to have lower uncertainty across all performance indicators, but it produced less accurate results, with the DR-KLT also showing greater inaccuracy than the IRF. In Cases 5 and 7, both the IRF and PRF demonstrated high accuracy, and their integration through DR-KLT yielded the most accurate results.

To verify the impact of time-varying uncertainty, we additionally applied a conventional Kalman filter using a single variance value estimated from the entire dataset. In this case, the PRF was statistically evaluated as having lower uncertainty due to its overall higher accuracy. As a result, the Kalman filter output leaned more heavily toward the PRF, and the results were nearly identical to those obtained using PRF alone. However, this approach consistently showed lower accuracy than the proposed DR-KLT, particularly in cases where the IRF temporarily outperformed the PRF or where performance fluctuated between frames.

The overall average results across simulations indicate that the proposed method achieves the highest accuracy and demonstrates stable, reliable performance under diverse environmental conditions.

4.2 | Lab-scale test

To verify the effectiveness of the proposed method on videos capturing the vibrations of an actual structure, a load was applied to a model structure, and the resulting vibrations were recorded. To validate the accuracy of the video-based measurements, an LVDT (linear variable differential transformer) was used to obtain reference vibration data. The video was recorded at a frame rate of 59 FPS, and the experimental setup is illustrated in Figure 10. The input video for the lab-scale experiment had a resolution of 1920 × 1080 pixels (Full HD).

The measured displacement results are shown in Figure 11, and Table 4 presents the average performance indicators and the RMSE of displacement measurements for each method. As seen in the graph, significant errors occurred between 14 and 17 s when using the PRF. This was caused by a temporary outlier, where one or more feature points briefly deviated significantly from the object due to tracking instability. In this case, the error was triggered by an obstacle temporarily moving in front of the object, causing partial occlusion and mis-tracking of some

FIGURE 10 Lab-scale test configuration.

TABLE 4 RMSE (root mean square error) of the error estimation model.

Corner tendency (%)		Bi-directional error (pixel)		Number of feature points (count)	
IRF	PRF	IRF	PRF	IRF	PRF
0.61	0.67	0.01	6.58	84.07	93.11

Magnitude of optical flow (pixel)		RMSE (pixel)		
IRF	PRF	IRF	PRF	DR-KLT
0.02	0.08	0.40	6.19	0.40

feature points. However, because reliable feature points near the object remained during this period, the PRF estimation naturally recovered after the outlier lost influence. This phenomenon was not due to fusion or correction but arose from the inherent multi-point averaging and filtering mechanism of the PRF calculation. During the same interval, the PRF showed sharp increases in bi-directional error and optical flow magnitude, indicating heightened uncertainty. In contrast, the IRF maintained stable displacement tracking, and the proposed method, by adaptively assigning greater weight to IRF, achieved both stability and accuracy through confidence-based fusion.

These results demonstrate that the proposed method can reliably achieve accurate results, even when either IRF or PRF yields inaccurate outcomes under certain conditions. When either IRF or PRF produces highly inaccurate results, the proposed method evaluates the uncertainty of the erroneous data as high. Through this process, the proposed method reduces the weight of inaccurate data and enables the stable and accurate estimation of displacement.

4.3 | On-site test

To assess the practical applicability of the proposed method, an on-site experiment was performed on a

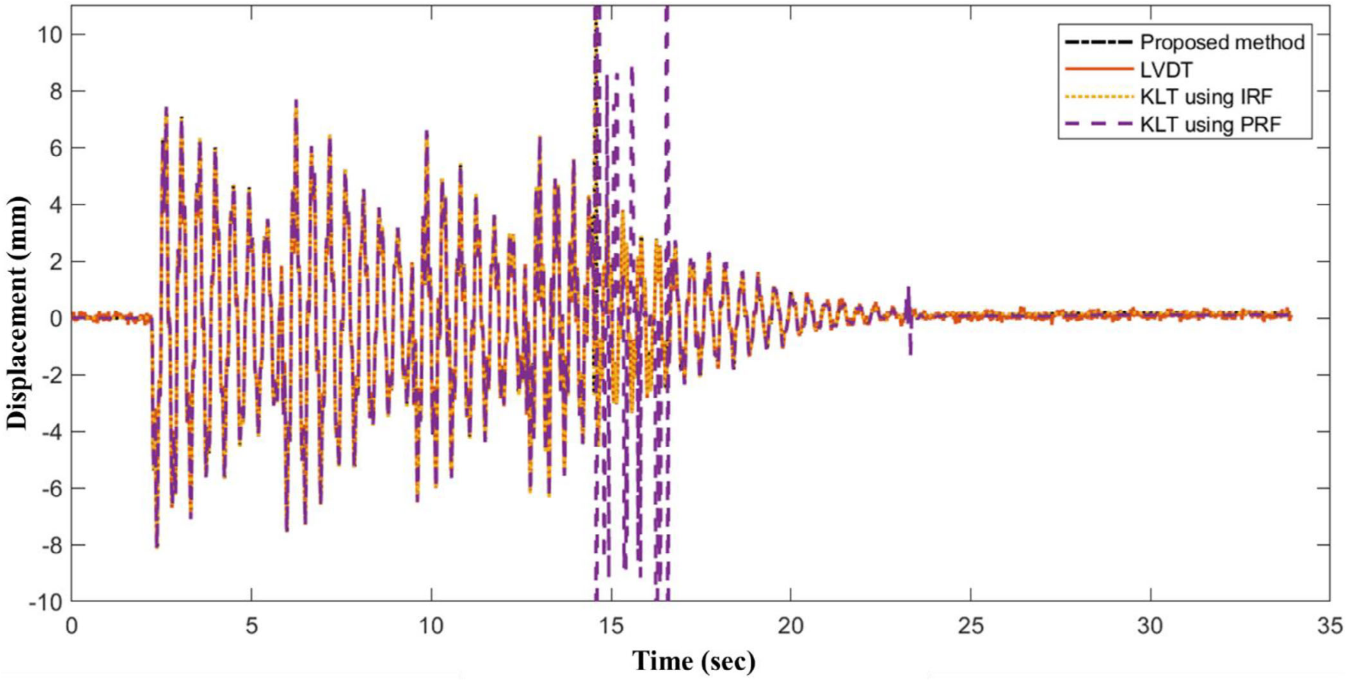

FIGURE 11 Estimated displacement using proposed DR-KLT, KLT using IRF, and KLT using PRF.

FIGURE 12 Overview of the on-site validation test.

pedestrian suspension bridge. The target bridge, located in Korea, is the Yangban-gil Suspension Bridge, which has a total length of 167 m and features a typical suspension bridge design. The pedestrian suspension bridge was selected due to its potential for significant displacement, making it suitable for displacement observation. During the experiment, vibrations were induced by manually applying a load at the center of the bridge, and the resulting motion was recorded on video. Figure 12 illustrates the pedestrian suspension bridge and the corresponding experimental setup. The on-site video was recorded at a resolution of 1920×1080 pixels.

In the acquired video, displacements were measured using both IRF and PRF, and the proposed method was employed to estimate a single fused displacement (Figure 13). During the measurement process, outliers were observed at specific time steps, notably at 5 and 18 s, in the data measured by either IRF or PRF. Despite the presence of these outliers, the proposed method successfully mitigated their impact, ensuring more stable results. The robustness of the method is demonstrated by its ability to utilize inlier data from either IRF or PRF to minimize the influence of outliers, leading to more stable and reliable outcomes. The presence of outliers at certain time steps highlights the inherent challenges in video-based displacement measurements, especially when environmental noise or slight variations in measurement conditions are present. The ability of the proposed method to reduce the impact of these outliers is crucial for ensuring the accuracy and reliability of the data, particularly in practical applications where external factors may introduce variability. By integrating IRF and PRF data, the method improves the consistency of displacement measurements, which is critical for accurate SHM. Future analyses could focus on examining the specific conditions that led to the presence of outliers, including changes in lighting, camera setup vibrations, or other environmental factors that could influence measurement accuracy. Moreover, investigating the sensitivity of the proposed method to varying levels of noise and outlier occurrences could offer deeper insights

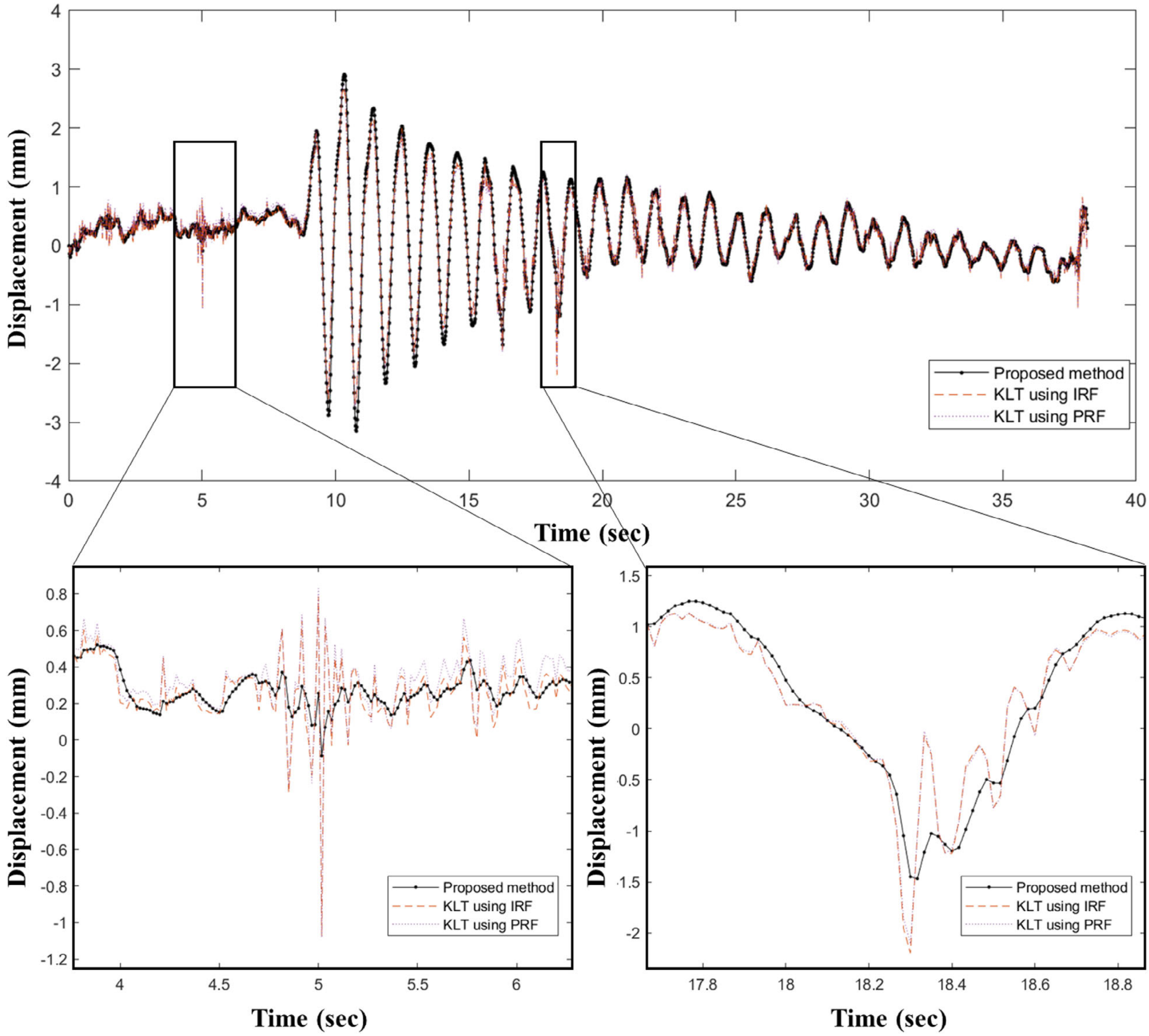

FIGURE 13 Measured displacement for the on-site validation test.

into its robustness and potential limitations, guiding future improvements and refinements.

5 | CONCLUSIONS

This paper presents a method for measuring displacement by utilizing two reference image frames and fusing the resulting displacement datasets through a time-varying Kalman filter. Performance indicators influencing the accuracy of the KLT tracker, including corner tendency, the number of feature points, bi-directional error, and the magnitude of optical flow, were analyzed. Based on these indicators, a model was developed to estimate the measurement errors of the KLT tracker. These estimated errors served as indicators of uncertainty in the measurement data for each time step within the time-varying Kalman filter.

The proposed method successfully estimated a single fused displacement from the dual-reference frames. The effectiveness of the method was validated through simulations, lab-scale experiments, and on-site tests. Simulation results demonstrated that the proposed method provides greater stability and accuracy compared to using either IRF or PRF alone. Specifically, the proposed DR-KLT method achieved an RMSE of 24.32 pixels, which was lower than that of PRF (24.75 pixels) and IRF (28.64 pixels), demonstrating its superior tracking accuracy. Additionally, the lab-scale tests showed that the proposed method yielded results closely matching those obtained from LVDT sensors, even when using real camera footage. Finally, the on-site tests verified that the proposed method can be reliably applied to actual structures, yielding consistent and stable results.

The DR-KLT approach explicitly overcomes the limitations of traditional IRF- and PRF-based tracking. While IRF struggles with large displacements and PRF accumulates drift over time, DR-KLT adaptively fuses both references based on real-time tracking confidence. This allows it to maintain robust and accurate displacement estimation across a wide range of conditions. Furthermore, its training-free design and lightweight implementation enhance its practicality for long-term SHM.

Future work could expand beyond limiting the reference frame to either the initial frame or the previous frame, potentially incorporating calculations and optimizations across all frames to identify the most suitable displacement scenario. Such a system could significantly improve the accuracy and reliability of displacement measurements, particularly in environments with varying external conditions.

These advancements could make substantial contributions to the field of SHM by providing engineers with more reliable tools for assessing the integrity and safety of critical infrastructure. The findings of this research lay the groundwork for future developments, with the proposed method expected to play a vital role in advancing the state of the art in SHM.

6 | DISCUSSION

This section provides further reflection on the strengths and limitations of the proposed DR-KLT method, along with comparisons to existing approaches and considerations for future research and implementation.

Although the proposed method is based on improving the traditional KLT tracker using dual-reference error fusion, it is also important to consider how it compares to other vision-based displacement measurement techniques. Digital image correlation (DIC) offers full-field displacement analysis but often requires high-contrast surface patterns, high-resolution images, and substantial computational resources. Dense optical flow methods can track detailed motion at the pixel level but are sensitive to image noise and lighting changes. Deep learning-based regression models can provide high accuracy but generally require large amounts of training data and suffer from poor interpretability. In contrast, the DR-KLT method requires no training, is computationally lightweight, and dynamically adapts to tracking confidence through time-varying Kalman filtering, making it more suitable for practical and scalable SHM applications.

Recent deep learning-based and hybrid vision-based methods have also been explored for structural displacement measurement (Pan et al., 2023; Perez-Ramirez et al., 2019). These approaches often achieve high accuracy but come with trade-offs in terms of complexity, computational load, and data requirements. Deep learning models typically require large, labeled datasets for training and may lack interpretability in real-world applications. Hybrid methods, which combine learning-based models with conventional trackers, provide more flexibility but remain sensitive to domain shifts and still depend heavily on pretraining. In contrast, the DR-KLT method operates without the need for pretraining and relies solely on readily available KLT performance indicators. It dynamically adapts to changes in tracking quality using a lightweight regression model and time-varying Kalman filter. This makes it more suitable for deployment in practical SHM scenarios, especially in environments with limited data, computing resources, or network access.

A known limitation of KLT-based trackers, including DR-KLT, is their dependency on the visibility and stability of feature points. In low-light or high-motion-blur environments, reliable corner detection may fail, degrading performance. Future improvements may involve integrating adaptive contrast enhancement, noise suppression, or combining DR-KLT with low-light-robust feature detectors such as learned descriptors or thermal-infrared sensors.

Although the proposed error estimation model improves the confidence-aware fusion process, its performance may be sensitive to the quality of data input. Low image resolution, motion blur, or unstable lighting can reduce the reliability of tracking indicators such as corner strength or optical flow magnitude. Furthermore, the regression model used for error estimation may introduce latency or computational cost, which could limit its use in edge-based or real-time monitoring systems. Although the error estimation model was trained using simulation data designed to cover a wide range of realistic conditions, including variations in lighting, motion, and image quality, we acknowledge that extreme or unforeseen environments may still fall outside the trained domain. As a result, future work will explore expanding the training dataset and incorporating online learning strategies to improve generalizability and robustness under such conditions. Future work could explore model simplification, feature selection, or online learning strategies to reduce computational burden while maintaining robustness.

In addition to vision-based tracking performance, it is important to consider the influence of the structure's dynamic characteristics on the measured results. For instance, higher vibration modes and lower damping ratios can cause faster motion, which increases motion blur and reduces corner stability. The distribution of feature points may also align with mode shapes, affecting local tracking accuracy. Future work will aim to establish a more direct relationship between vision-based tracking performance and structural mechanics to enhance the physical interpretability of measurements.

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